SASKEN

Next Phase of Android

Having established itself as a secure, robust, stable platform, how is Android gearing up for proliferation into newer segments?

Whitepaper

Abstract:

As of 2020, Android has achieved widespread adoption in the devices segment, with a staggering 75% market share. Its mainstays and handhelds have been designed to reach out to various end-user touch points. Be it an infotainment unit in cockpit systems, television systems, enterprise rugged devices, point of sale terminals, digital signage, and so on. In the last decade, with a clear focus on establishing itself as a robust, secure, and flexible platform along with a huge developer base, Android has been steadily embraced by various segments. This paper discusses this evolution, focus shifts, and the next anticipated move of the Android platform.

Author:

Narasimha Bashyam, Senior Architect

Sasken Technologies Ltd.

Table of Content

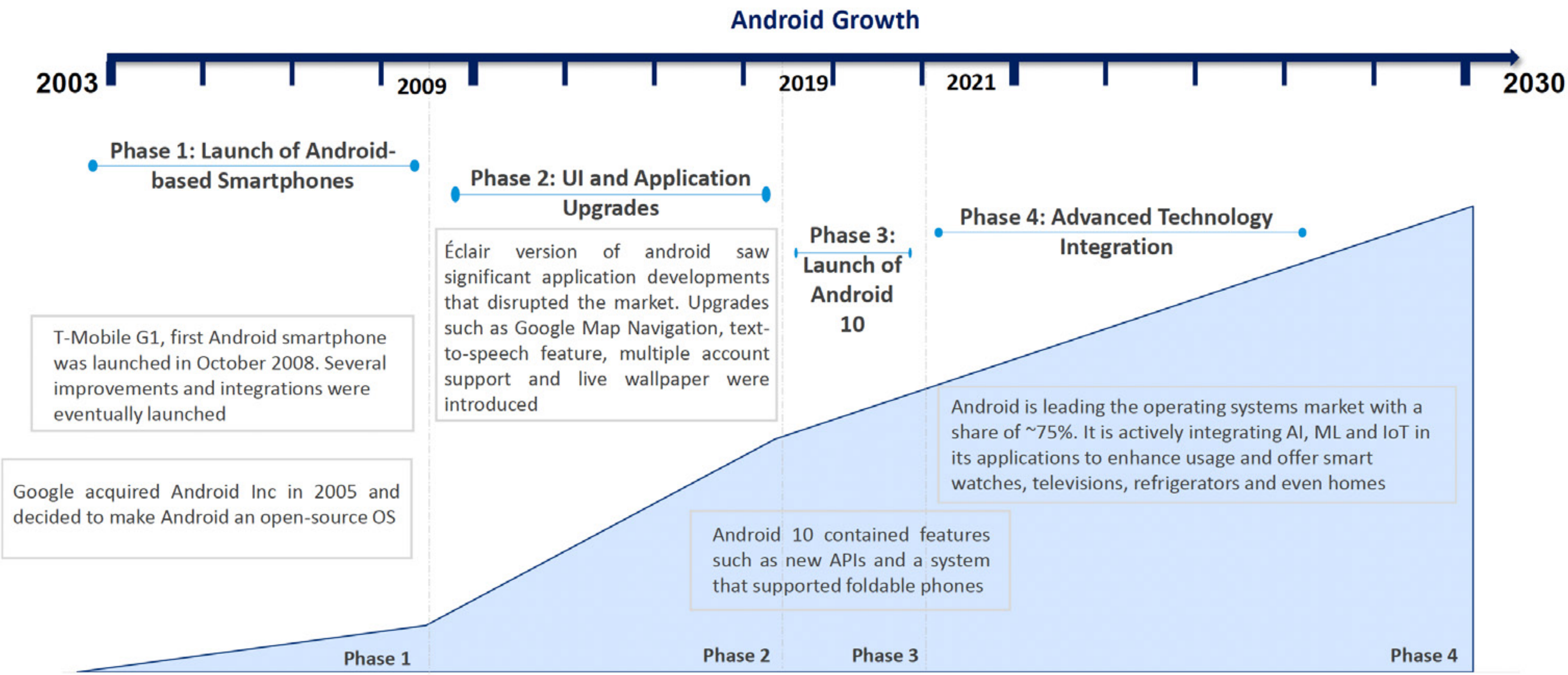
Introduction

With the advent of smartphones in early 2000, Apple and Google took different approaches to capturing market share. Apple sought to create a distinct user experience in the design of the iPod and moved toward touch-based displays. Touch-based displays were a significant milestone in the evolution of smartphones. Google quickly stepped in with its acquisition of Android Inc. in 2005.

Google took the open-source route, which ensured a quick and huge following of developers. Its focus on enabling tools, compliance suites paved the way for a maturing platform. Although fragmentation had been a concern in the early days, Google successfully handled it with stringent compliance rules. The quicker turnarounds of releases via pre-access programs with chipset vendors also significantly moved OEMs towards later Android versions.

With significant strides in security engineering via Selinux, hardware-backed keystores, and Treble architecture, Android closed the gaps with security and paved the way for efficient customization management. Enterprise customers started embracing Android after a significant focus on security engineering.

Android 10 introduced new multiscreen dimensions and addressed some critical OEM needs, including vehicle HAL, custom launcher, and system UI. Its usage is further diversified across segments like smart watches, televisions, refrigerators, and connected homes with the integration of AI, ML, and IOT.



Source: Industry blogs, Research Papers, Whitepapers, and Draup Team Analysis
Note: The above chart is indicative only

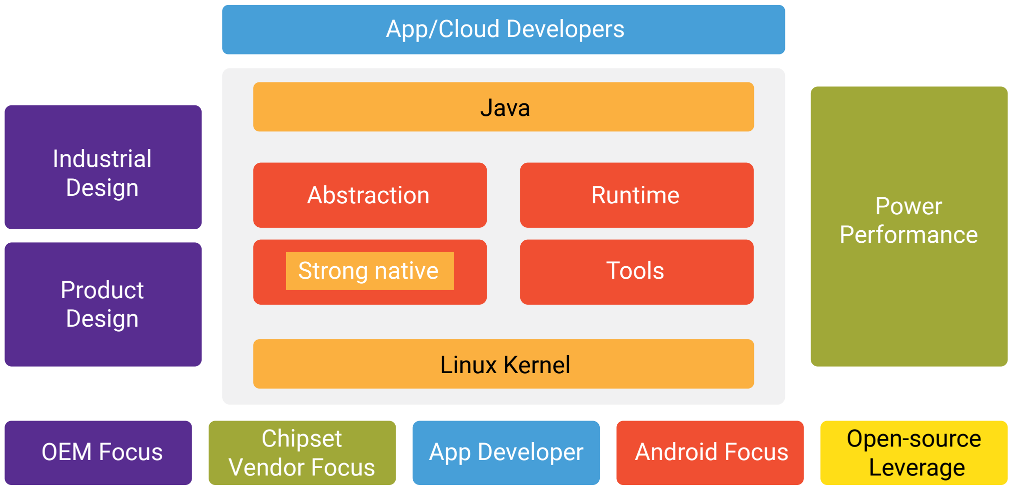
1. OS: Operating System

Source: Draup

Android Approach vs. Linux vs. QNX

Android adoption by various industries is primarily attributed to

- Ease of Application Development
- Powerful platform/subsystem libraries
- Pre-access partnership with chipset vendors
- Ease of customization by OEMs



While OEMs can focus on industrial and product design, chipset vendors focus on power and performance. Being an open-source platform, chipset vendors have a say in blending their architecture with Android abstraction. Android continues to focus on strengthening the run-time, decoupling architecture, and tools.

Significantly App developers are totally abstracted on the nuances of the underlying HAL architecture and can focus on realizing the user-stories.

For instance, in the automotive sector as compared to other OS, Android scores on

- The Tools support - Emulator, Studio, Profilers, Debuggers
- OEM control and ease of customization
- Application Ecosystem
- Quicker Time to Market

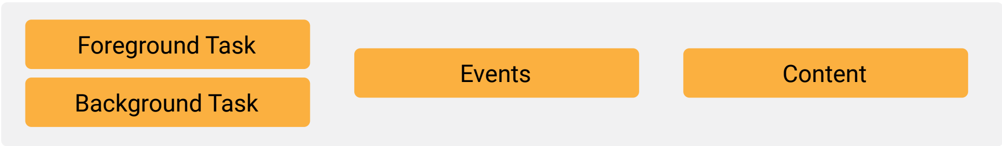
Factors	Comparative score	Android	Automotive Grade Linux	QNX
Time to Market	⬆	• TTM has been reduced by using a stable operating system with each Android version. • Good emulator support helps in app development before a prototype is available.	• TTM benefits from unified code base distribution.	• Stable OS, less legal compliance, and OEM independence help TTM
SW Stack comprehensiveness	⬆	• Good Tech Stack Support • Abstraction robust	• SW dev kits provided • Application templates for easy usability	• Pre-Integrated UI Framework • Virtual Machine Support
OEM Control	⬆	• Highly customizable UI • AOSP source code access to customize ensuring compatibility	• Unified code base distribution	• OEM branding UI can be independently developed
OEM Cost	⬆	• AOSP-free, Google apps must be licensed. • Costs associated with security connectivity interfaces	• Platform free but maintenance, security, app dev cost is high	• High cost of ownership

Security	→	• Application sandboxing • Mandatory Access Control via SeLinux • File-based encryption	• Lower safety as opposed to the MAC	• Good security control over connectivity interfaces
Functional Safety	↓	• No ASIL compliance	• NO ASIL compliance	• ASIL-D compliance
Application Ecosystem	↑	• Large application developer base • Access to Google Apps • There are numerous third-party apps available on the Play Store	• OEM responsibility to develop apps • Minimal in-built apps	• Third-party multimedia and Web apps can be run • No built-in apps
OTA updates	→	• Frequent OTA updates help keep you up to date on features. • At the same time, increases the safety, status quo risks • Doesn’t require a third-party solution	• Primary Update for OEM Applications	• OEMs have control over the frequency of updates

Similarly Android scores better in most of the segments when compared to the other OS in their respective segments.

Android – Where it stands

Android Abstraction is inextricably linked with User Journeys. The user journey is the series of actions that an end-user takes while performing a task on the device. The abstraction revolves around

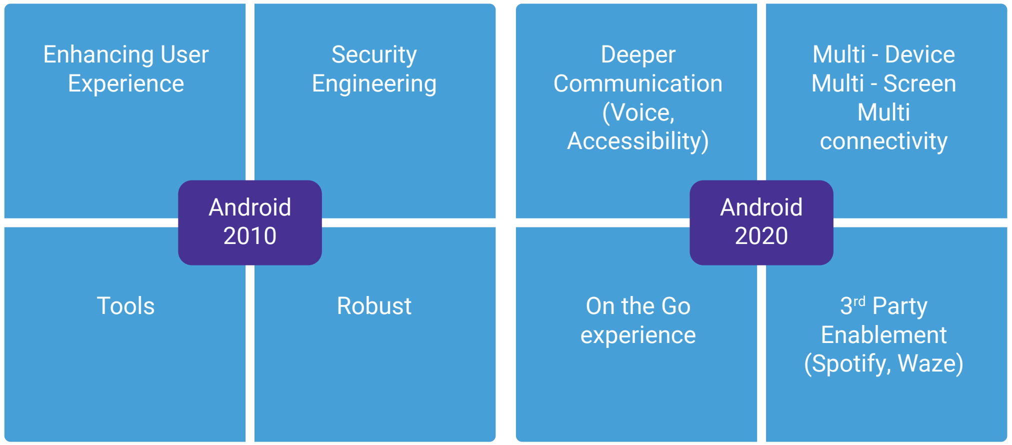


As it started, Android was focusing on enhancing the user experience, security engineering, and tools. Given its solid abstraction design and user journey focus, end-users found navigation to be more intuitive.

Because of the security enhancements in Android KitKat, the focus shifted to "Material Design" UI/UX. With Material Design, applications are more engaging, immersive, and communicative. Android started moving towards split-screen, separating OS system components from vendor components via the Treble Architecture.

As of 2020, having established itself as a secure, robust, and feature-rich OS, the focus started moving towards multi-devices, deeper communication, and an on-the-go experience for end-users. The intention is to be always-on with the end-user in the form of hand-held devices, wearables, TVs, and vehicle infotainment systems, either at home or at the workplace.

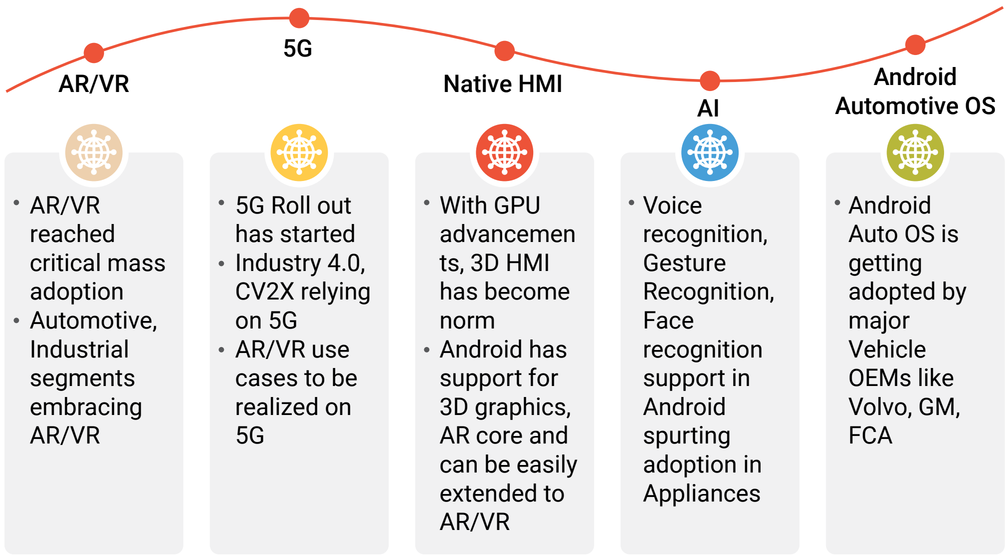
Android is also enabling support for 3rd-party applications, accepted by end users in various geographies, for non-device segments. Ensuring necessary guidelines to meet the segment-specific operational requirements, say, "avoiding driver distraction," is a key step in this regard.



The Next Move

The next wave of technology has begun. Artificial intelligence has already proliferated into various segments, be it hand-held, autonomous driver assistance, or warehouse inventory management. Gestures and voice assistants are new forms of human-machine interaction. Machine-to-machine interactions are becoming a reality with low-latency, high-bandwidth 5G networks. Immersive experiences are being tried and tested. Digital twins and virtual worlds are the new paradigms.

Android is embracing these new technologies in its sphere of influence. ARCore is already part of the AOSP to enable augmented reality. Native libraries are enhanced to enable Depth API functionality and motion detection. AI/ML kit libraries can be easily integrated with AOSP. For 3D and VR capabilities, the Unity SDK and Unreal Engine SDK are also available.



HMI is taking on different dimensions: immersive, 3D, voice, and gestures are all integral parts of it. 5G deployment is enabling AR and VR to become mainstream and offload high-performance computing onto the cloud.

These new drivers—5G, AR/VR, AI, voice, and gestures—are bringing in new user experiences and complex use cases that span across multiple devices. Android, with its new focus on multi-devices and multi-screens, should cater to these demands.

Android needs to enhance its subsystems to cater to these technological changes. Only then can the vision of "On-the-Go," "Deeper Communication," and "Multi-Devices" be achieved.

Particularly when M2M communication to leapfrog with advent of 5G, thin versions of Android for various kind of devices similar to Android Go Edition – may be a necessity.

Soon it may be a case of a single "Android Super Box" that caters to the needs and demands of multiple-form-factor devices at home. The devices enter "slave mode" once at home to save power, and the "Android Super Box" serves as the master.

Conclusion

Android has started moving into a new era of embracing advanced technologies. The AOSP is resilient to the new technologies' induced enhancements. The focus again shifts to new UI/UX immersive experiences, security engineering for interconnected devices, simulator tools, and robust operating systems.

Sasken Solution Engineering is driving the accelerators that help customers quickly embrace this focus shift and quicker time-to-market of next-generation products.

About the Author

Narasimha Bashyam, a Senior Architect in the Android space, has more than 20 years of experience working on various Android platforms across the Consumer Electronics, Automotive, Enterprise, and Semiconductor segments providing thought leadership to major OEMs and Tier-1 customers. His expertise includes Android IVI systems, System Integration, Platform Security, Multimedia Frameworks, and Android-for-Work. He aligns himself with business teams towards delivering offerings, solutions, GTM strategy, and competition analysis.

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 32 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.

SASKEN

Next Phase of Android

Having established itself as a secure, robust, stable platform, how is Android gearing up for proliferation into newer segments?

marketing@sasken.com | www.sasken.com

USA | FINLAND | GERMANY | JAPAN | INDIA

October 2022